Qubit Electronics-Formula Sheet

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1 Qubit Introduction

Fundamental Qubit Representation and State Vectors

- **General Superposition:** $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ where α and β are complex probability amplitudes.
- **Normalization Condition:** $|\alpha|^2 + |\beta|^2 = 1 = \langle\psi|\psi\rangle$
- **Hermitian Conjugate:** $|\psi\rangle^\dagger = \langle\psi| = [\alpha^*, \beta^*]$
- **Multi-Qubit System Scaling:** A system of N qubits scales to 2^N components. For $N = 3$: $|\psi\rangle = c_1|000\rangle + c_2|001\rangle + \cdots + c_8|111\rangle$

The Bloch Sphere Geometry

- **General State Formula:** $|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$
- **Bloch Vector Coordinates:** $|0\rangle$ (North Pole): $(0, 0, 1)$; $|1\rangle$ (South Pole): $(0, 0, -1)$; $|+\rangle$: $(1, 0, 0)$; $|-\rangle$: $(-1, 0, 0)$

Quantum Measurement and the Born Rule

- **State Collapse:** Measurement forces $|\psi\rangle$ to collapse into either $|0\rangle$ or $|1\rangle$.
- **Born Rule:** $P(|a\rangle) = |\langle\psi|a\rangle|^2$

Basis States in Different Representations

- **Computational Basis:** $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
- **Hadamard Basis:** $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$
- **Circular Basis:** $|+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$, $|-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$

Single-Qubit Unitary Operators

- **X (NOT):** $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
- **Y:** $\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
- **Z:** $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
- **Hadamard (H):** $\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

- **Phase (S):** $\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
- **T:** $\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
- **Rotation Gates:** $R_x(\theta) = e^{-i\theta X/2}, \quad R_y(\theta) = e^{-i\theta Y/2}, \quad R_z(\theta) = e^{-i\theta Z/2}$

Two-Qubit Interactions: CNOT Gate

- **Mapping Rule:** $|A\rangle|B\rangle \rightarrow |A\rangle|A \oplus B\rangle$
- **Matrix Representation:** $\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

Density Matrix Formalism

- **Definition:** $\rho = |\psi\rangle\langle\psi|$
- **Mixed State Ensemble:** $\widehat{\rho} \equiv \sum_i p_i |\Psi_i\rangle\langle\Psi_i|$ where p_i are probabilities and $\sum_i p_i = 1$
- **Trace Property:** $\text{Tr}(\rho^2) = 1$ if pure state;< 1 if mixed state

Qubit Performance Metrics

- **Coherence Time:** $\frac{1}{T_2} = \frac{1}{T_\phi} + \frac{1}{2T_1}$
- **Gate Capacity:** $N_{\text{gates}} \sim \frac{T_2}{t_{\text{gate}}}$

2 Superconducting Qubits

Fundamental Superconductivity and Cooper Pairs

- **Collective Wave Function:** $\psi = \psi_0 \exp\left[-\mathrm{j}\frac{1}{\hbar}(Et - \mathbf{p} \cdot \mathbf{x})\right]$ where $\mathrm{j} = \sqrt{-1}$
- **Cooper Pair Density:** $\rho = \psi \cdot \psi^* = |\psi_0|^2$
- **Meissner Effect:** $B = 0$

Josephson Junction Dynamics

- **Phase Difference:** $\phi = \varphi_2 - \varphi_1$
- **Current-Phase Relation:** $I = I_C \sin\phi$
- **Voltage-Phase Relation:** $V = \frac{\Phi_0}{2\pi}\dot{\phi}$ where $\Phi_0 = \frac{h}{2e} = \frac{2\pi\hbar}{2e}$
- **Josephson Inductance:** $L = \frac{\Phi_0}{2\pi I_C \cos\phi}$ and $L_j = \frac{\Phi_0}{2\pi I_C}$
- **Josephson Energy:** $E_j(1 - \cos\phi)$

LC Circuit Quantization

- **Hamiltonian:** $H = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}$
- **Resonant Frequency:** $\omega_{01} = \frac{1}{\sqrt{LC}}$
- **Canonical Commutation:** $[\hat{\Phi}, \hat{Q}] = \mathrm{i}\hbar$ and $[\hat{a}, \hat{a}^\dagger] = 1$

Transmon Qubit Parameters

- **Non-linear Inductance:** $L(\Phi) = \frac{\Phi_0}{2\pi I_C \cos(2\pi\Phi/\Phi_0)}$
- **Transition Energy:** $E_{01} = \hbar\omega_{01} \approx 10$ GHz
- **Anharmonicity:** $E_{01} \neq E_{12}$ allows selective qubit control
- **Operating Temperature:** ≈ 10 mK, requiring $k_B T \ll \hbar\omega_{01}$
- **Gate Time:** $T_\pi \approx 4$–20 ns
- **Coherence Times:** $T_1, T_2 \sim 100$–1000 μs

3 Circuit QED

Fundamental Light-Matter Interaction

- **Dipole Coupling:** $H = -\vec{d} \cdot \vec{E}(t)$ where $\vec{d}$ is the electric dipole moment
- **Rabi Frequency:** $\Omega_R = \frac{dE_0}{\hbar}$ for classical field amplitude E_0

Resonator Physics

- **Fabry-Pérot Resonance:** $2L = m\lambda_m$ where m is the mode number
- **Mode Separation:** $\Delta f = \frac{c}{2L}$
- **LC Resonator Frequency:** $\omega_r = \frac{1}{\sqrt{L_r C_r}}$
- **Quality Factor:** $Q = \frac{\omega_r}{\kappa}$ where κ is the decay rate

Jaynes-Cummings Model

- **Full Hamiltonian:** $H = \hbar\omega_r \hat{a}^\dagger \hat{a} + \frac{\hbar\omega_{01}}{2}\hat{\sigma}_z + \hbar g(\hat{a}\hat{\sigma}_+ + \hat{a}^\dagger \hat{\sigma}_-)$
- **Rotating Wave Approximation (RWA):** Neglects rapidly oscillating terms at frequency $\omega_r + \omega_{01}$
- **Jaynes-Cummings (RWA):** $H_{JC} = \hbar\omega_r \hat{a}^\dagger \hat{a} + \frac{\hbar\omega_{01}}{2}\hat{\sigma}_z + \hbar g(\hat{a}\hat{\sigma}_+ + \hat{a}^\dagger \hat{\sigma}_-)$

Coupling Strength

- **Coupling Formula:** $g = \omega_r \frac{C_g}{C}\left(\frac{E_j}{2E_c}\right)^{1/4}\sqrt{\frac{\pi Z_r}{R_k}}$ where C_g is gate capacitance, Z_r is resonator impedance, and $R_k = \frac{h}{e^2} \approx 25.8\ \mathrm{k}\Omega$
- **Typical Range:** $g/2\pi \approx 100$ MHz to 1 GHz (strong coupling regime)

Dispersive Regime

- **Condition:** $|\Delta| = |\omega_{01} - \omega_r| \gg g$
- **Dispersive Shift (AC Stark Shift):** $\chi = \frac{g^2}{\Delta}$
- **Effective Hamiltonian:** $H \approx (\omega_r + \chi\hat{\sigma}_z)\hat{a}^\dagger \hat{a} + \frac{\omega_{01}}{2}\hat{\sigma}_z$
- **State-Dependent Resonator Frequency:** $\omega_r^{(0)} = \omega_r - \chi$ (ground state $|0\rangle$); $\omega_r^{(1)} = \omega_r + \chi$ (excited state $|1\rangle$)
- **Readout via Dispersive Shift:** Phase accumulation $\propto \chi\hat{\sigma}_z t$ enables qubit-state discrimination without excitation

Circuit QED Implementation

- **Cavity Analogue:** Superconducting coplanar waveguide (CPW) replaces optical Fabry-Pérot mirrors
- **Artificial Atom:** Transmon qubit (Duffing oscillator) coupled to resonator via capacitive coupling C_g
- **Substrate:** Aluminum or niobium on sapphire for low loss
- **Field Confinement:** Tightly confined to $\sim 10^{-3}\lambda$, enabling large vacuum fluctuations $E_0 \approx 0.2$ V/m
- **Coupling Enhancement:** Circuit QED achieves $\approx 2 \times 10^4\times$ stronger coupling than cavity QED

System Hamiltonian Hierarchy

- **Full Circuit Ham.:** $H = \frac{Q_r^2}{2C_r} + \frac{\Phi_r^2}{2L_r} + 4E_c\hat{n}^2 - E_j \cos\hat{\phi} - g(\hat{b}^\dagger - \hat{b})(\hat{a}^\dagger - \hat{a})$
- **Two-Level Approx.:** $H = \omega_r \hat{a}^\dagger \hat{a} + \frac{\omega_{01}}{2}\hat{\sigma}_z - g(\hat{\sigma}_+ + \hat{\sigma}_-)(\hat{a}^\dagger + \hat{a})$
- **After RWA:** $H = \omega_r \hat{a}^\dagger \hat{a} + \frac{\omega_{01}}{2}\hat{\sigma}_z + g(\hat{a}^\dagger \hat{\sigma}_- + \hat{a}\hat{\sigma}_+)$

Transmon-Specific Considerations

- **Anharmonicity:** $\delta = (E_{12} - E_{01})/\hbar$ from Duffing nonlinearity; enables selective single-qubit addressing
- **Flux Tunability:** SQUID integration allows $\omega_{01}(\Phi_{\text{ext}})$ tuning via external magnetic flux
- **Charging Energy:** $E_c = \frac{e^2}{2C}$ determines qubit-cavity detuning range
- **Josephson Energy:** E_j sets nonlinearity strength; transmon regime requires $E_j \gg E_c$

4 Qubit Control

Readout and the Dispersive Regime

- **Full Jaynes-Cummings Hamiltonian:** $\hat{H} = \omega_r \hat{a}^\dagger \hat{a} + \frac{\omega_q}{2}\hat{\sigma}_z - g(\hat{a}^\dagger \hat{\sigma}_- + \hat{a}\hat{\sigma}_+)$
- **Dispersive Approximation:** $\hat{H} \approx (\omega_r + \chi\hat{\sigma}_z)\hat{a}^\dagger \hat{a} + \frac{\omega_q}{2}\hat{\sigma}_z$
- **Dispersive Regime Condition:** $g \ll |\omega_q - \omega_r|$, where $\Delta = \omega_q - \omega_r \gg g, \kappa$
- **Purcell Decay Rate:** $\gamma_p = \left(\frac{g}{\Delta}\right)^2 \kappa$
- **Cavity Decay Rate:** $\kappa = \frac{\omega_r}{Q}$
- **Optimal Readout Balance:** $\chi/\kappa \approx 1$
- **Purcell Filter:** Frequency-selective element with $\kappa_{\text{eff}}(\omega_q) \ll \kappa(\omega_r)$

Transmon Hamiltonian and Microwave Drive

- **Circuit Hamiltonian:** $\hat{H} = \frac{\hat{Q}^2}{2C_\Sigma} + \frac{\hat{\Phi}^2}{2L} + \frac{C_d}{C_\Sigma}V_d(t)\hat{Q}$
- **Charge Operator:** $\hat{Q} = -iQ_{\text{zpf}}(\hat{a} - \hat{a}^\dagger)$ where $Q_{\text{zpf}} = \sqrt{\frac{\hbar}{2Z_0}}$
- **Characteristic Impedance:** $Z_0 = \sqrt{\frac{L}{C_\Sigma}}$
- **Two-Level Truncation:** $\hat{a} \to \hat{\sigma}_-$, $\hat{a}^\dagger \to \hat{\sigma}_+$
- **Truncated Qubit Hamiltonian:** $\hat{H} = -\frac{\omega_q}{2}\hat{\sigma}_z + \Omega V_d(t)\hat{\sigma}_y$ where $\Omega = \frac{C_d}{C_\Sigma}Q_{\text{zpf}}$

Rotating Frame and Rotating Wave Approximation

- **Unitary Transformation:** $U(t) = e^{i\omega_q t\hat{\sigma}_z/2}$
- **Monochromatic Drive:** $V_d(t) = V_0 s(t)(I\sin(\omega_d t) - Q\cos(\omega_d t))$
- **Effective Drive Hamiltonian (RWA):** $\hat{H}_d' = -\frac{\Omega}{2}V_0 s(t)(I\hat{\sigma}_x + Q\hat{\sigma}_y)$
- **RWA Validity:** Neglects rapidly oscillating terms at $\omega_d + \omega_q$; valid when $\omega_d \approx \omega_q$

Virtual Z Gates and Phase Control

- **Software Phase Update:** $I' = I\cos\phi - Q\sin\phi$; $Q' = I\sin\phi + Q\cos\phi$
- **Example (Virtual Z, $\phi = \pi/2$):** Initial pulse $(I = 1, Q = 0) \to (I' = 0, Q' = 1)$ converts X-rotation to Y-rotation

5 Qubit Characterization

Bloch Sphere and State Representation

- **Qubit State:** $|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$
- **Bloch Vector:** $\vec{r} = \begin{pmatrix} r_x \\ r_y \\ r_z \end{pmatrix} = \begin{pmatrix} \sin\theta\cos\phi \\ \sin\theta\sin\phi \\ \cos\theta \end{pmatrix}$

Region	Axis Designation	Physical Counterpart
North Pole	$+z$	Longitudinal Axis
South Pole	$-z$	Longitudinal Axis
Equator	x-y Plane	Transverse Plane
Precession	$\phi(t)$	Rotation at ω_{01} (Lab Frame)

Reference Frames and Resonance

- **Laboratory Frame:** The physical frame where the Bloch vector precesses around the z-axis at the qubit transition frequency ω_{01} when a transverse component exists.
- **Rotating Frame:** A reference frame rotating at frequency ω_r. When $\omega_r = \omega_{01}$, the Bloch vector appears stationary.
- **Resonance Condition:** On resonance in the rotating frame, the ideal interaction Hamiltonian H' is zero. Any observed movement in this frame is due to drive fields or noise.

Decoherence and Relaxation Rates

- **Energy Relaxation (T_1):** $\Gamma_1 \equiv 1/T_1$
- **Rate Synthesis:** $\Gamma_1 = \Gamma_{1\downarrow} + \Gamma_{1\uparrow}$
- **Boltzmann Excitation:** $\Gamma_{1\uparrow} = \Gamma_{1\downarrow} e^{-\hbar\omega/k_B T}$
- **Pure Dephasing:** Γ_ϕ arises from longitudinal noise coupling via the z-axis and causes frequency fluctuations $\delta\omega(t)$.
- **Transverse Relaxation (T_2^*):** $\Gamma_2 = \frac{\Gamma_1}{2} + \Gamma_\phi$ or $\frac{1}{T_2^*} = \frac{1}{2T_1} + \frac{1}{T_\phi}$
- **Phase-Breaking Justification:** Energy relaxation is phase-breaking; once $|1\rangle \to |0\rangle$, phase information is destroyed.
- **Technology Note:** Transmon qubits are generally T_1-limited and typically exhibit relatively large T_2 times.

Characterization Measurement Protocols

- **Time of Flight (TOF):** Calibrates the propagation delay between readout pulse generation and signal arrival at the acquisition IP within the RFSoC.
- **Requirement:** Mandatory first step; applied to all subsequent measurements.
- **Resonator Spectroscopy:** Sweeps the readout tone to find the fundamental frequency; high power is used to decouple the qubit from the resonator.
- **Drive Spectroscopy:** Two-tone experiment used to calibrate the drive frequency to the qubit transition frequency f_q.
- **Calibration Example:** Identified $f_q = 3774.16$ MHz.
- **Rabi Oscillations:** Used to calibrate frequency, duration, and gain after identifying f_q.
- **Calibration Example (Digital Gain):** Calibrated π-pulse: 0.054; calibrated $\pi/2$-pulse: 0.027.
- **Resonator Punchout:** 2D sweep of readout gain versus frequency to map power response and identify the dressed readout frequency.
- **Physics:** At high power, the Josephson junction nonlinearity is suppressed; as power decreases, the resonance shifts to the dressed/dispersive readout frequency.

Coherence Measurements

- T_1 **Measurement:** $f(t) \propto e^{-t/T_1}$; signal decay of 63% occurs at $t = T_1$.
- **Ramsey Protocol (T_2^*):** $X_{\pi/2}$ — wait τ — $X_{\pi/2}$ — measure.
- **Mechanism:** Drive is detuned by $\delta\omega$ to induce controlled precession.
- **Spin Echo (T_2):** $X_{\pi/2}$ — wait $\tau/2$ — Y_π — wait $\tau/2$ — $X_{\pi/2}$ — measure.
- **Purpose:** The central π-pulse compensates for quasi-static noise.
- **Interpretation:** T_2 (echo) $> T_2^*$ (Ramsey) indicates low-frequency noise.

6 TLine

Transmission Line Parameters

- **Resistance per unit length:** R_0 [Ω/m]
- **Inductance per unit length:** L_0 [H/m]
- **Capacitance per unit length:** C_0 [F/m]
- **Conductance per unit length:** G_0 [S/m]

The Telegrapher Equations

- **Voltage equation:**
$$\frac{\partial V(z,t)}{\partial z} = -R_0\, i(z,t) - L_0 \frac{\partial i(z,t)}{\partial t}$$
- **Current equation:**
$$\frac{\partial i(z,t)}{\partial z} = -G_0\, V(z,t) - C_0 \frac{\partial V(z,t)}{\partial t}$$
- **Lossless line:** for $R_0 = 0$ and $G_0 = 0$,
$$\frac{\partial^2 V}{\partial z^2} = L_0 C_0 \frac{\partial^2 V}{\partial t^2}$$

Wave Propagation Dynamics

- **Propagation speed:**
$$v = \frac{1}{\sqrt{L_0 C_0}}$$
- **Voltage solution:**
$$V(z,t) = V^+(t - z/v) + V^-(t + z/v)$$
- **Characteristic impedance:**
$$Z_0 = \sqrt{\frac{L_0}{C_0}}$$
- **Current waves:**
$$i^+(z,t) = \frac{V^+(z,t)}{Z_0}, \qquad i^-(z,t) = -\frac{V^-(z,t)}{Z_0}$$

Reflections and Terminations

- **Voltage reflection coefficient:**
$$\rho_v = \Gamma_L = \frac{R_L - Z_0}{R_L + Z_0}$$
- **Current reflection coefficient:**
$$\rho_i = -\rho_v = \frac{Z_0 - R_L}{Z_0 + R_L}$$
- **Matched load:** if $R_L = Z_0$, then $\Gamma_L = 0$ and no reflections occur.

Power on the Transmission Line

- **Instantaneous power:**
$$p(z,t) = V(z,t)\, i(z,t)$$
- **Average power for a traveling wave:**
$$P_{\text{avg}} = \frac{V_{\text{rms}}^2}{Z_0} = I_{\text{rms}}^2 Z_0$$
- **Power transfer:** maximum power is delivered when the line is matched to the load.

7 Smatrix

Scattering Parameters for an Impedance Discontinuity

- **Impedances:** Two semi-infinite lines meet at a junction with characteristic impedances Z_1 and Z_2.
- **Power-normalized waves:**
$$a_n = \frac{V_n^+}{\sqrt{Z_n}}, \qquad b_n = \frac{V_n^-}{\sqrt{Z_n}}$$
- **Port convention:** Port 1 corresponds to Z_1, and Port 2 corresponds to Z_2.

Reflection and Transmission Coefficients

- **Reflection at the junction from Port 2:**
$$S_{22} = \Gamma_{A^+} = \frac{Z_1 - Z_2}{Z_1 + Z_2}$$
- **Voltage transmission coefficient:**
$$\frac{V_A^-}{V_A^+} = \frac{2Z_1}{Z_1 + Z_2}$$
- **Power-normalized transmission:**
$$S_{12} = \sqrt{\frac{Z_2}{Z_1}} \cdot \frac{V_A^-}{V_A^+} = \frac{2\sqrt{Z_1 Z_2}}{Z_1 + Z_2}$$
- **Symmetric result:** by reciprocity,
$$S_{21} = \frac{2\sqrt{Z_1 Z_2}}{Z_1 + Z_2}$$
- **Reflection seen from Port 1:**
$$S_{11} = \frac{Z_2 - Z_1}{Z_1 + Z_2}$$

Scattering Matrix

$$\mathbf{S} = \frac{1}{Z_1 + Z_2}\begin{pmatrix} Z_2 - Z_1 & 2\sqrt{Z_1 Z_2} \\ 2\sqrt{Z_1 Z_2} & Z_1 - Z_2 \end{pmatrix}$$

Device Properties

- **Reciprocal:** $S_{12} = S_{21}$.
- **Not symmetric:** generally $S_{11} \neq S_{22}$ unless $Z_1 = Z_2$.
- **Lossless junction:** the S-matrix is unitary, $\mathbf{S}\mathbf{S}^\dagger = \mathbf{I}$.

Fundamental Power Wave Definitions

- **Incident wave:**
$$a_i = \frac{V_i + Z_{ri} I_i}{2\sqrt{Z_{ri}}}$$
- **Reflected wave:**
$$b_i = \frac{V_i - Z_{ri} I_i}{2\sqrt{Z_{ri}}}$$
- **Power normalization:** $|a_i|^2$ and $|b_i|^2$ represent power in watts.
- **Net real power at port i:**
$$P_i = |a_i|^2 - |b_i|^2$$
- **Spatial propagation:**
$$a(x) = a(0)e^{-jkx}, \qquad b(x) = b(0)e^{+jkx}$$
- **Reference impedance:** Z_{ri} is the characteristic impedance used to normalize port waves.

General N-Port S-Matrix Framework

- **Black-box model:** internal circuit details are replaced by input-output relations between incident and reflected waves.
- **S-matrix relation:**
$$\mathbf{b} = \mathbf{S}\mathbf{a}$$
- **Matrix size:** for an N-port network, $\mathbf{S}$ is an $N \times N$ matrix.
- **Scattering parameter definition:**
$$S_{ij} = \left.\frac{b_i}{a_j}\right|_{a_k=0 \text{ for all } k\neq j}$$
- **Matched-load condition:** $a_k = 0$ is realized by terminating all non-driven ports with their reference impedances.

Two-Port Network

- **Equations:**
$$b_1 = S_{11}a_1 + S_{12}a_2, \qquad b_2 = S_{21}a_1 + S_{22}a_2$$
- **Matrix form:**
$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

Physical Interpretation of S-Parameters

- S_{ii}**:** reflection coefficient seen looking into port i with all other ports matched.
- S_{ij}**:** transmission coefficient from port j to port i.
- S_{21}**:** forward transmission (gain/loss).
- S_{12}**:** reverse isolation.
- **Power directionality:** S-parameters quantify how much power is reflected or transmitted through the network.

8 VNA

Power Wave Variable Definitions

- a_1**:** incident wave at Port 1.
- b_1**:** power reflected from Port 1.
- b_2**:** power transmitted through the DUT and leaving Port 2.
- a_2**:** incident wave at Port 2.

Directional Coupler Performance

- **Coupling factor (dB):**

$$\text{Coupling (dB)} = -10\log\left(\frac{P_{\text{coupling forward}}}{P_{\text{incident}}}\right)$$

- **Isolation factor (dB):**

$$\text{Isolation (dB)} = -10\log\left(\frac{P_{\text{coupled reverse}}}{P_{\text{incident}}}\right)$$

- **Directivity (dB):**

$$\text{Directivity (dB)} = \text{Isolation (dB)} - \text{Coupling (dB)}$$

- **Mainline loss:** total insertion loss in the primary through-line.

VNA Measurement Signal Relationships

- **Working principle:** a 3-way resistive splitter is used in simplified models to sample the input signal.
- **Instrument architecture:** actual VNA hardware uses directional couplers to separate incident and reflected waves.
- **Coherent receivers:** RX REF1, RX TEST1, RX REF2, and RX TEST2 share a common reference oscillator.

S-Parameter Extractions

- **Forward measurements:** when SW1 is in position 1,

$$S_{11} = \frac{b_1}{a_1}, \qquad S_{21} = \frac{b_2}{a_1}$$

- **Reverse measurements:** when SW1 is in position 2,

$$S_{22} = \frac{b_2}{a_2}, \qquad S_{12} = \frac{b_1}{a_2}$$

- **2×2 S-matrix:**

$$\begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}$$

- **Index convention:** i = destination/output port, j = source/input port.

Laboratory Reference

- **Core function:** characterizes amplitude and phase response over frequency without knowing the DUT topology.
- **Applications:** filters, amplifiers, antennas, resonators, and couplers.
- **Systems engineering:** with antennas, the VNA can function as a radar system.
- **Advanced diagnostics:** can support material characterization and imaging applications.

9 Modulations

Reference Nomenclature and Variable Definitions

- f_s**:** DAC/ADC sampling rate.
- f_{rfmax}**:** upper bound of the allocated RF band.
- f_c**:** RF center frequency (carrier frequency).
- f_{C-IF}**:** local oscillator frequency for IF translation.
- f_{IF}**:** intermediate frequency.
- $f_{s(BB)}$**:** baseband sampling rate.
- **ADC:** analogue-to-digital converter.
- **DAC:** digital-to-analogue converter.
- **DSP:** digital signal processing.

Nyquist Sampling Criterion for RF Systems

- **Primary condition:**

$$f_s > 2f_{rfmax}$$

- **Condition met:** the digital/analogue interface can be placed directly at the RF stage.
- **Condition not met:** a staged architecture using an intermediate frequency f_{IF} is required.

Architectural Definitions

- **Heterodyning:** direct translation between baseband and RF in a single step.
- **Superheterodyning:** multi-stage translation using an intermediate frequency f_{IF}.
- **Stage 1:** baseband to intermediate frequency.
- **Stage 2:** intermediate frequency to radio frequency.

Comparative Analysis of Radio Architectures

- **Direct RF:** ADC/DAC at the RF stage near the antenna; requires very high sampling rates.
- **IF Sampling:** digital processing occurs at f_{IF}, while analogue mixers and LOs perform the RF translation.
- **Baseband Sampling:** the interface is placed at baseband, and modulation/demodulation occurs in analogue hardware.
- **Interface position logic:** higher sampling rates move the digital/analogue interface closer to the antenna.

Technical Advantages of Digital Implementation

- **Operational precision:** digital logic provides deterministic and accurate operation.
- **Hardware robustness:** reduced drift from tolerances, thermal effects, and ageing.
- **System efficiency:** smaller footprint, simpler BOM, and lower total power consumption.
- **Flexibility:** reconfigurable hardware supports multiple communication standards via software.

Summary Reference Table

Architecture	**Typical Sampling Rate**	**Interface Position**
Direct RF	$\approx$ 10 GSps	RF stage near antenna
IF Sampling	10s MSps	Intermediate frequency f_{IF}
Baseband Sampling	10s–100s kSps	Baseband stage

10 Heterodyning

Fundamental Sampling Criterion

- **Nyquist-Shannon condition:**

$$f_s > 2f_{rfmax}$$

- f_s**:** sampling rate of the ADC or DAC.
- f_{rfmax}**:** maximum frequency component in the RF-modulated signal.
- **Consequence:** if the condition is not met, an intermediate frequency stage is required.

Modulation and Demodulation Schemes

- **Heterodyning:** single-stage translation between baseband and RF using carrier frequency f_c.
- **Superheterodyning:** two-stage translation via intermediate frequency f_{IF}.
- **Stage 1:** baseband → IF.
- **Stage 2:** IF → RF.
- **Typical IF band:** 10s of MHz to 100s of MHz.
- **Typical RF band:** 100s of MHz to 10s of GHz.

Comparative Analysis of Radio Architectures

Architecture	Interface Placement	Typical Sampling Rate	Logic
Direct RF	RF stage	$\approx$ 10 GSps	DUC/DDC with interpolators, decimators, a
IF Sampling	IF stage	10s MSps	Analog IF-to-RF translation with image reje
Baseband Sampling	Baseband stage	10s–100s kSps	Analog-domain modulation/demodulation

Digital Processing Advantages

- **Operational precision:** deterministic accuracy and reduced nonlinear errors.
- **Long-term reliability:** reduced sensitivity to drift and aging effects.
- **Physical and fiscal efficiency:** smaller footprint, simpler BOM, and lower power.
- **Architectural versatility:** reconfigurable hardware for multiple waveforms and bands.

RFSoC Integration

- **RFSoC:** integrates high-performance ADCs/DACs, digital logic, and processor cores on one CMOS die.
- **Benefit:** enables direct digital modulation with reduced analog complexity.